United International **University (UIU)**

Department of Computer Science and Engineering

CSE 1325: DIGITAL LOGIC DESIGN, Midterm Summer 2025

Total Marks: **30,** Duration: 1 hour 30 minutes

[Any examinee found adopting unfair means including copy from another examinee will be expelled from the trimester/program as per UIU disciplinary rules.]

Answer <u>All</u> Questions

1. A) Convert $(1432)_7$ and $(1180)_9$ to BCD (Binary Coded Decimal) format. Perform the BCD addition of the two numbers and show all necessary steps. **[3]**

B) Find the values of the radix a and b from the following equations: **[3]**

$$\frac{(57)_a + (37)_b}{(112)_4} = \frac{49}{22}$$
$$\frac{(81)_b + (27)_a}{(221)_6} = \frac{56}{85}$$

2. A) Prove the identity of the following Boolean equation using algebraic manipulation: **[3]**

$$\bar{A}B(\bar{D} + \bar{C}D) + B(A + \bar{A}CD) = B$$

B) Find the complement of the following function **[3]**

$$F\,(X, Y, Z) = (\bar{X} + Y + \bar{Z})(X + \bar{Y})(X + Z)$$

3. For the following Boolean function F, use K-Map And find the following:

$$F\,(a.b.c.d) = \left(\bar{a}bd + a\bar{c}\bar{d} + abd + ac\bar{d}\right)$$

A. Convert it to canonical form (SOM). **[1]**
B. Simplified sum-of-products (SOP) and **[2]**
C. Simplified product-of-sums (POS) form. **[2]**
D. Between minimized SOP and POS, which form will you choose to implement the function and why? **[1]**

4. Consider the following Boolean function:

$$F(A, B, C, D) = \textstyle\sum_m(0, 1, 4, 6, 7, 9, 11, 13, 14) + \sum_d(3, 5, 8)$$

A. Find all the Prime Implicants and **[2]**
B. determine which of the Prime Implicants are Essential. **[2]**
C. Find the optimal solution using selection rule. **[2]**

5. Consider a smart smoothie maker that checks the taste of a smoothie depending on the fruits added. The system checks the presence of four types of fruits: mangoes (M), watermelons (W), lemons (L), and passion fruits (P). Mango and watermelon are sweet fruits, while lemon and passion fruit are sour. If the value of a variable is **1** that means the fruit is present in the smoothie and **0** means the fruit is absent from the smoothie. The smoothie is considered **sweet (output 1)** if there are more sweet fruits (mangoes and watermelons) than sour fruits (lemons and passion fruits). It is considered **sour (output 0)** if the number of sour fruits is greater than or equal to the number of sweet fruits. The output is **don't care (X)** if there are no fruits present at all.

Design a Boolean function **F** that can be used to make a circuit to check these conditions in the smoothie maker.

You have to

A. Draw the truth table. **[2]**

B. Find the simplified expression for the output bit in Sum-of-Product (SOP) form using K-map. **[3]**

C. Draw the circuit diagram using basic gates. **[1]**

Example:

- Input 1000 → Output = 1 (Mango only → sweet majority)
- Input 0011 → Output = 0 (Lemon + Passion fruit → sour majority)
- Input 0110 → Output = 0 (Watermelon + Lemon → equal number of sweet and sour fruits)
- Input 0000 → Output = X (No fruits → Don't care)